DagShield

Land-record conflict screening with evidence you can check

Live demo
dagshield.newslensbd.com
Scan for the one-click judge demo

LandTech 2026 | Track: Digital Land Records, Verification & Digitization | **Problem P3: Catch Fraud & Double-Selling**

Team stratifyX Global

1 Chosen problem

P3: Catch Fraud & Double-Selling. Spot the same land sold to more than one buyer, fake or altered documents, and records that do not match between registration and mutation. DagShield also covers part of P2 (verify deeds with hashing, a digital signature and a QR code) and part of P5 (a public “check before you buy” page).

2 Why it matters in Bangladesh

As the LandTech brief says, many land disputes start with fake deeds, with old records that do not match each other (CS, SA, RS and BS khatians), and with records kept in separate offices that do not talk to each other. A deed is registered at the sub-registry office, but the owner changes in the khatian only after mutation at the AC Land office. In between, the registration record and the khatian name different people. That gap is where a seller can sell the same plot again.

Today a registrar usually sees one document at a time, and a buyer often sees nothing at all. A viewer that only shows records does not reveal how records relate to each other, so a second sale of the same land, or a seller who is not the owner, can pass unnoticed until it reaches court.

5
transparent, versioned rules

3 colours
green, yellow, red, each with a written reason

2 languages
Bangla and English, same results

0 AI
models at run time, so every answer can be reproduced

3 Our solution: DagShield

DagShield links parcel lineage, ownership entitlements, deeds, transfers and mutation entries in one model. Deterministic rules flag **potential** conflicts and prove each flag with the record IDs it used and the arithmetic. It serves three users:

Sub-registrar

Checks a deed **before** it is registered. The answer is green (registered), yellow (registered, flagged for review) or red (blocked), with the reason in plain words.

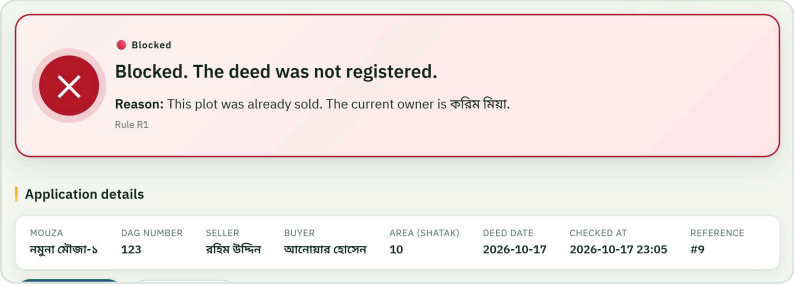
Citizen or buyer

Picks a mouza, types a dag number and an owner name, and learns whether that person is the recorded owner and whether the mutation is done. No login.

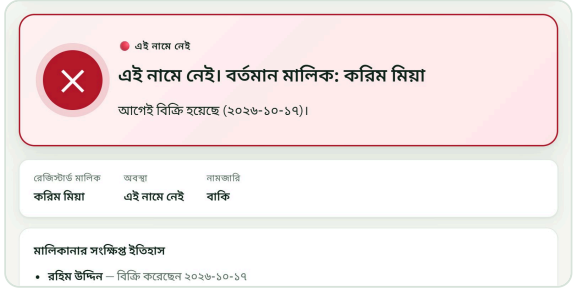
Reviewer or judge

Opens an alert and sees the rule, the evidence records and the calculation. Signed deeds with a QR code show exactly which field was changed in a copy.

What it looks like (live prototype, synthetic data)

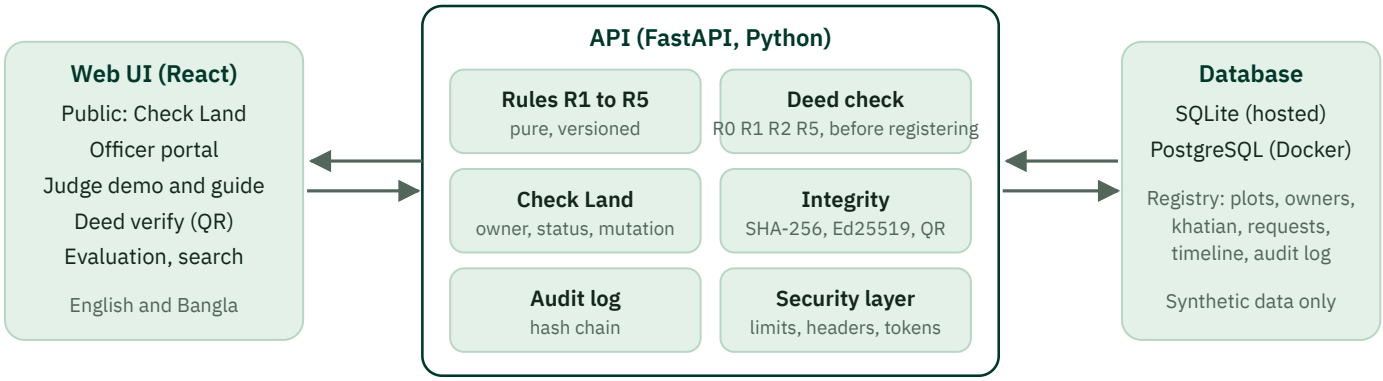


Officer portal: a second sale of plot 123 is blocked, rule R1, with the reason and the current owner.



Check Land in Bangla: the earlier owner is told “not under this name”, with the sale date.

4 How it works: architecture and tech stack



A deed is checked before it is registered

1. The officer enters mouza, dag, seller, buyer, area and date.
2. The checker reads who holds the plot now and the earlier transfers. It works on plain data, so it is easy to test.
3. It answers with a colour, a rule and a reason in Bangla and English. A blocked request changes nothing.
4. Green and yellow move ownership. The request, the result and the timeline are saved, and an entry is added to the hash-chained audit log.
5. The public page reads the same data, so the new owner appears at once. The khatian changes only when the mutation is completed, so a fresh owner shows amber until then.

Tech stack

Layer	Choice
Web UI	React 18, TypeScript, Vite; Bangla and English
API	Python 3.12, FastAPI, Pydantic (strict input checks)
Data	SQLAlchemy; SQLite on the hosted demo, PostgreSQL in Docker
Integrity	SHA-256, Ed25519 signature, QR code, hash-chained log
Security	Rate limits, strict CSP headers, signed session tokens, scrypt password hash, secrets only in the environment
Quality	102 automated tests, lint with security rules, CI, secret and synthetic-data scanners
Hosting	One Python app on cPanel; Docker Compose for local use

The rules

Rule	What it finds	Shown to the user as
R0	The mouza and dag are not in the record	🔴 Red: not found in supplied dataset
R1	The same portion sold again, or a plot that was already sold	🔴 Red: current owner named
R2	The seller is not the owner, or sells more than the record holds	🔴 Red: record shows X, request is Y
R3	A registered deed with no matching mutation, or the reverse	🟡 Review: not found in supplied dataset
R4	Child plots that do not add up to the parent, or an uncertain split	🟡 Review: difference shown
R5	Another sale within 30 days of the last one (a heuristic, never a finding)	🟡 Yellow: registered, flagged

Wording is strict: “potential conflict”, “not found in supplied dataset”, “human review needed”. DagShield never says “fraud confirmed” or “ownership verified”. Tests enforce this.

Evidence that it works (on our synthetic fixtures)

Ten labelled scenarios plus integrity cases run from code: **14 of 14 cases pass**, with 10 true positives, 0 false positives and 0 false negatives, and identical results on repeated runs. This is a small synthetic set, so it shows that the rules behave as designed, not how they would perform on real records. An Evaluation page in the app recomputes these numbers on demand.

5 Main features

- **Officer portal:** login, dashboard counts, new-deed form with a green / yellow / red check and the reason, plot history timeline, alert list, and a form to add plots (names in Bangla or English).
- **Check Land (public):** a green, amber or red answer in Bangla or English. It shows only the owner, status, mutation status and earlier owners, never alerts, scores or officer names.
- **Mutation-aware:** registration and khatian are tracked separately, so “mutation pending” is visible to buyers and officers.
- **Deed integrity:** every synthetic deed is hashed and signed; a QR code leads to a verify page that names the changed field in a tampered copy.
- **Evidence for every alert:** rule and version, record IDs, calculation, explanation and a suggested next check.
- **Judge demo:** one click starts a guided 12-step tour with sample data and the expected result of each step.

6 How it could be used across the country

DagShield is a screening aid that keeps people in charge. A phased path, each step with a gate:

Phase 1	**Pilot in one upazila** with the land authorities. Gate: written authorisation and a data-sharing agreement.
Phase 2	**Read-only link** to registration and mutation records through an authorised API or scheduled export. Gate: security review.
Phase 3	**Portion identity:** derive the plot-portion key from the deed’s area and boundaries and the mutation split instead of the synthetic key. Gate: measured match rate against records reconciled by hand.
Phase 4	**Shadow mode:** run the rules beside the current process and tune them with reviewers. Gate: an agreed false-alarm rate.
Phase 5	**Signing and scale-out:** key custody by the registration authority (hardware-protected), a legal opinion on digital signatures, then upazila by upazila with offline QR checks. Always: role-based access, audit logging, personal-data protection, reviewer training.

Who benefits

User	What DagShield does for them
Sub-registry offices	Checks the seller’s ownership and earlier sales before a deed is registered.
AC Land and Union Land offices	Shows mismatches between registration and mutation, and pending mutations.
Banks and developers	Confirms the recorded owner before accepting land as collateral or buying it.
Citizens and buyers	“Check before you buy”, with no login.

7 Limits we state openly

- All data is made up. It cannot show readiness for real records, which need authorisation first.
- A deed forged at the source and then signed passes the QR check, so rules R1 to R4 are the second line of defence.
- Names are matched by spelling (case and spaces ignored). “Rahim Uddin” and “রহিম উদ্দিন” are different names to the program.
- Planned, not in the demo: owner login, real SMS and OTP, NID checks, map and GIS checks, and OCR of scanned deeds.

8 Team, mentors and declarations

Team stratifyX Global
Students: Gulam Sakaria (Team Leader), Md Abdur Rahim
Faculty mentor: **Nushrat Jahan Oyshi**, Lecturer, CSE
Industry mentor: **Shafin Ahmed**, CSE-TECH
Live demo: dagshield.newslensbd.com Judge demo: dagshield.newslensbd.com/judge Code: github.com/gulamsakaria/dagshield

Declarations. Synthetic data only: no real names, NID numbers, phone numbers or land records, nothing scraped from government sites. Open-source tools and licences are listed in the README. AI tools used during development (Claude) are disclosed in the README; no AI model runs inside the product. Fonts: IBM Plex Sans and Noto Sans Bengali (SIL OFL). Map outlines: geoBoundaries (CC0).